WORKED SOLUTION

STEP I 2013, Question 9

Mechanics • Projectiles and Collisions

STEP I 2013, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

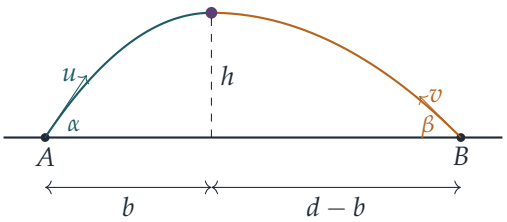
step.maths.org/questions/13-s1-q9

The key idea

At the top of its flight a projectile has no vertical velocity, so after a collision there the time to fall back down equals the time it took to rise. To land back where it started, each particle must therefore retrace its horizontal journey in the same time, which forces its horizontal velocity simply to be reversed. Put that into conservation of momentum, and use the fact that the two particles reach the top at the same instant, and everything follows.

Solution

We model A and B as particles moving freely under gravity, with no air resistance, and the impulse in the collision acts along the line of centres. Since the collision is direct and both particles are moving horizontally at that moment, this impulse is horizontal. Take the x-axis along the plane from A's starting point towards B's, so B starts at $x = d$.



Meeting at the tops

A has vertical velocity $u\sin\alpha - gt$, so it reaches its greatest height at time $t = u\sin\alpha/g$. Likewise B is highest at $t = v\sin\beta/g$. They were projected simultaneously and collide when each is at its top, so these times are equal:

$$T = \frac{u\sin\alpha}{g} = \frac{v\sin\beta}{g}, \qquad \text{so} \qquad u\sin\alpha = v\sin\beta. \tag{1}$$

Just before the collision, A has velocity $u\cos\alpha$ in the positive x-direction and B has velocity $v\cos\beta$ in the negative x-direction; neither has any vertical velocity.

After the collision

Ask yourself How can we say anything about the velocities after the collision when we know nothing about the coefficient of restitution?

The return condition does the work. The impulse is horizontal, so A still has zero vertical velocity just after the collision, at height h above the plane. Falling from rest (vertically) through h takes exactly as long as rising through h did, namely T. During that time A must move horizontally from $x = b$ back to $x = 0$, where $b = u\cos\alpha\, T$ is how far it travelled on the way up. So its new horizontal velocity is

$$-\frac{b}{T} = -u\cos\alpha\,:$$

A's horizontal velocity is exactly reversed, and in the same way so is B's. (In fact each particle retraces its own path backwards.)

Conservation of momentum in the x-direction now gives

$$mu\cos\alpha - Mv\cos\beta = -mu\cos\alpha + Mv\cos\beta, \qquad \text{so} \qquad mu\cos\alpha = Mv\cos\beta. \qquad (2)$$

In other words, the total momentum is zero both before and after. Dividing (2) by (1), which is allowed since α and β are strictly between 0 and $\frac{1}{2}\pi$ so both sides of (1) are non-zero,

$$m\cot\alpha = M\cot\beta,$$

as required.

Where the collision happens

In time T, A travels a horizontal distance $b = u\cos\alpha\, T$ and B travels $v\cos\beta\, T$. Together they cover the gap, so

$$u\cos\alpha\, T + v\cos\beta\, T = d.$$

By (2), $v\cos\beta = \dfrac{m}{M}\,u\cos\alpha$, so

$$b\left(1 + \frac{m}{M}\right) = d \qquad \Longrightarrow \qquad b = \frac{Md}{m+M},$$

as required. (The heavier B is, the slower it moves horizontally, and the further from A's start the meeting point lies.)

The height of the collision

Using $v^2 = u^2 + 2as$ vertically for A, the greatest height is

$$h = \frac{u^2\sin^2\alpha}{2g}, \qquad \text{while} \qquad b = u\cos\alpha\cdot\frac{u\sin\alpha}{g} = \frac{u^2\sin\alpha\cos\alpha}{g}.$$

Dividing,

$$\frac{h}{b} = \frac{\sin\alpha}{2\cos\alpha} = \tfrac{1}{2}\tan\alpha, \qquad \text{so} \qquad h = \tfrac{1}{2}\,b\tan\alpha.$$

Checks

- Symmetry: with $m = M$ we get $\alpha = \beta$ and $b = \frac{1}{2}d$, so the particles meet in the middle, as they must.
- The height formula says the line from A's start to the top of its path has gradient $\frac{1}{2}\tan\alpha$, half the launch gradient: the familiar fact that a parabola's tangent at the start meets the axis of symmetry at twice the height of the vertex. The same calculation for B gives $h = \frac{1}{2}(d-b)\tan\beta$, and this agrees with $\frac{1}{2}b\tan\alpha$ precisely because $m\cot\alpha = M\cot\beta$.
- Numerically, take $m = 1$, $M = 2$, $\alpha = 45°$, $u = 10\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$. Then $\cot\beta = \frac{1}{2}$, $v = \sqrt{50}/\sin\beta \approx 7.91\,\mathrm{m\,s^{-1}}$, and $d \approx 7.65\,\mathrm{m}$, giving $b = \frac{2}{3}d \approx 5.10\,\mathrm{m}$ and $h = \frac{1}{2}b \approx 2.55\,\mathrm{m}$. Stepping both motions forward in time by computer, with the horizontal velocities reversed at the collision, confirms that the particles meet at their tops at this point and both land back at their starting points.

Common slips

- Trying to use a coefficient of restitution: none is given, and none is needed, since the return condition fixes the velocities after impact.
- Forgetting that the particles must reach their tops at the same time, which gives (1); without it the cotangents cannot be produced.
- Sign errors in the momentum equation: B is moving in the negative direction before the collision.

Beyond STEP

The return condition amounts to time reversal: Newton's equations under gravity look the same if we run the clock backwards, so reversing a particle's velocity sends it back along its own path. Here the total momentum is zero, which means we are working in the centre-of-mass frame, and in that frame an elastic collision just reverses both velocities, exactly what happens here (so this collision is perfectly elastic). Centre-of-mass frames and conservation laws are explored beautifully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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